

DYNAMIC TRAFFIC FLOW MANAGEMENT

Submitted by

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**In partial fulfillment of the
requirements for the Final Project**

**Department of Electronics and
Communication Engineering
Academic Year 2025–2026**

CERTIFICATE

This is to certify that the project entitled “Dynamic Traffic Flow Management System” has been successfully carried out by Ashrith Patel in partial fulfillment of the requirements for the award of the degree in Electronics and Communication Engineering (ECE).

The work presented in this project report is a genuine and original work carried out under the guidance and supervision of the concerned faculty during the academic year 2025–2026. This project has been found satisfactory and approved for the submission of the project documentation.

ACKNOWLEDGEMENT

I express my sincere gratitude to our respected Principal, Head of the Department, and faculty members of the Department of Electronics and Communication Engineering for their valuable support and encouragement throughout the completion of this project.

I would like to convey my heartfelt thanks to my project guide for providing continuous guidance, motivation, and technical support during the development of the project titled “Dynamic Traffic Flow Management System.”

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Finally, I express my gratitude to my parents for their constant encouragement and support, which helped me successfully complete this project documentation.

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INTRODUCTION

- Traffic congestion is a persistent issue in urban areas around the world, leading to significant economic, environmental, and social impacts. As cities continue to grow, the management of traffic flow becomes increasingly challenging. Traditional traffic management systems often fall short in addressing the dynamic nature of urban traffic, which is influenced by various factors such as time of day, special events, accidents, and changing weather conditions.
- To tackle this problem and improve the overall quality of urban life, there is a growing need for innovative approaches to traffic flow management. This project, “Dynamic Traffic Flow Management for Sustainable Urban Mobility,” aims to explore and implement advanced solutions that leverage cutting-edge technology and data-driven strategies to optimize traffic flow in urban environment .
- When an emergency strikes, every second counts. Access for emergency vehicles, such as ambulances, can mean the difference between life and death.
- Our main aim is to give path for emergency vehicles .
- Let’s see how it possible .

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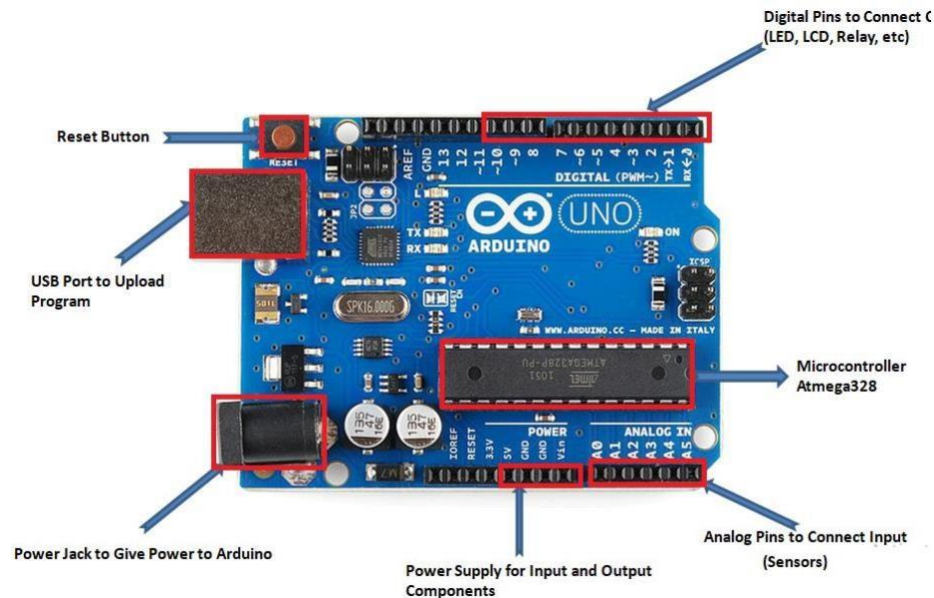
COMPONENTS

1. Arduino UNO
2. IR sensor
3. Servo motors
4. Bread Board
5. I2c module & LCD display
6. RFID

COMPONENTS

1.Arduino UNO:

- The word “uno” means “one” in Italian and was chosen to mark a major redesign of the Arduino hardware and software.
- Arduino is an open-source platform which is used for hardware and software .The Arduino Uno is a popular microcontroller board used for creating electronic projects. It's user-friendly and comes with digital and analog pins that allow you to control various hardware components. You can program it using the Arduino IDE, making it accessible for beginners and experienced makers. It's a versatile platform for building everything from simple LED projects to complex robotics and IoT applications



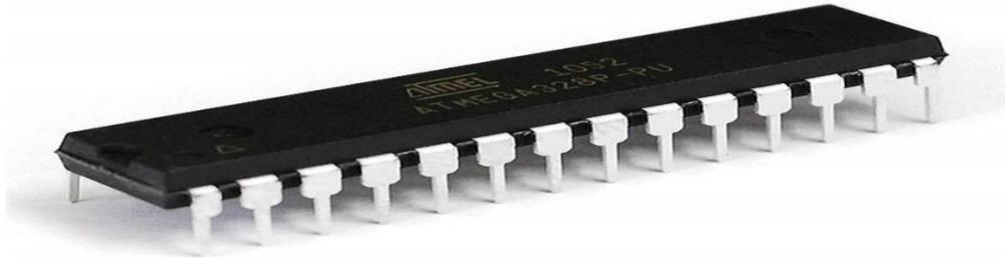
1.1 pictorial diagram of Arduino Uno Board

1.2. ATmega328 microcontroller:

The ATmega328 is the brains of the Arduino Uno. It executes the code that you upload to the board, making decisions and controlling various components connected to it.

Its an 8-bit microcontroller, meaning it processes data in 8-bit chunks.

- ATmega328 has 28 pins in total. It has 3 Ports in total which are named as Port B, Port C and Port



1.2.1 Atmega328 microcontroller

The ATmega328 typically operates at 16 MHz on the Arduino Uno, but it can run at different clock speeds depending on your configuration

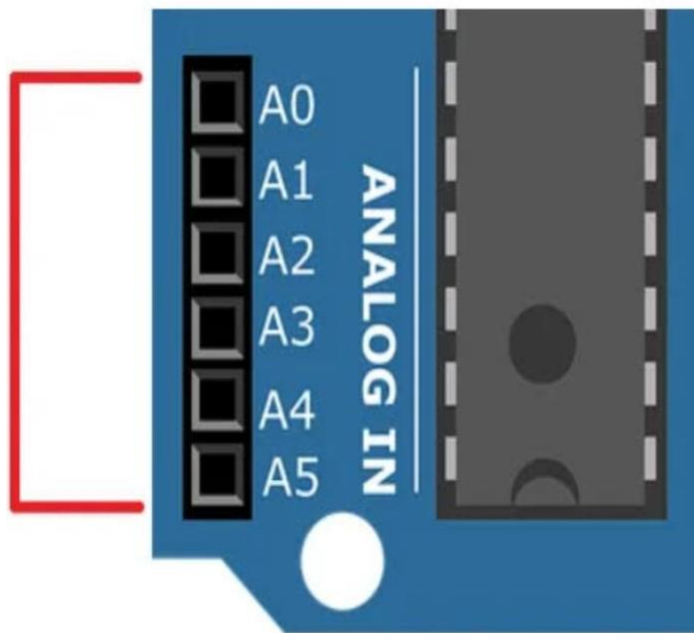
- **Flash Memory:** It has 32 KB of Flash memory, which is used to store the program that the microcontroller runs. In the case of the Arduino Uno, this is where your sketches (programs) are stored.
- **SRAM:** It has 2 KB of SRAM (Static Random-Access Memory), which is used for data storage during program execution.
- **EEPROM:** It also contains 1 KB of EEPROM (Electrically Erasable Programmable Read-Only Memory), which

provides non-volatile storage for data that needs to be retained across power cycles.

- **I/O Pins:** The ATmega328 has 23 general-purpose I/O pins, some of which support features like PWM, analog-to-digital conversion (ADC), and external interrupts.
- **Low Power Modes:** The microcontroller offers different low power modes, making it suitable for battery-powered applications.
- **Communication:** It supports serial communication through UART (Universal Asynchronous Receiver-Transmitter).
- **Peripherals:** It includes other features like SPI (Serial Peripheral Interface) and I2C (Inter-Integrated Circuit) for interfacing with various sensors and devices.
- **Operating Voltage:** It can operate at 5V or 3.3V, depending on the specific board design and configuration.
- **Programming:** The ATmega328 is typically programmed in C/C++ using the Arduino IDE or Atmel Studio.
- The ATmega328 is known for its reliability, ease of use, and a rich ecosystem of libraries and resources, making it a popular choice for hobbyists, students, and professionals for a wide range of embedded systems project.

1.3 Analog pins:

- The analog pins on the Arduino Uno are crucial for interfacing with analog sensors, measuring analog signals, and converting them into digital values for



processing within the microcontroller.

1.3.1 analog pins

- Many sensors and devices produce analog signals, such as temperature sensors, light sensors, potentiometers, and more. Analog pins allow the Arduino to read these signals accurately.

1.4 Power jack:

- The power jack is important to the Arduino Uno because it allows you to provide electrical power from an external source, like a battery or a wall adapter.



1.4.1 power jack in Arduino Uno

1.5 USB port:

- USB(universal serial bus), technology used to connect computers with peripheral devices.
- The USB port serves as a communication interface between the Arduino and a computer. It's used for uploading code (sketches) to the Arduino and for serial communication between the board and the computer.

1.6 Power Source: When the Arduino is connected to a computer via USB, it can also draw power from the USB connection, allowing it to be powered and operated directly from the computer.



1.6.1 power source in Arduino UNO

1.7. Reset button:

1.7.1. Program Restart: Pressing the reset button restarts the program uploaded to the Arduino. This can be useful for restarting the code without having to disconnect and reconnect the Arduino to your computer.



1.7.1 reset button in Arduino Uno

1.7.2. Entering Programming Mode: If you want to upload a new program to the Arduino, you typically press the reset button just before or during the upload process. This puts the Arduino into a mode where it's ready to receive new code.

1.7.3. Troubleshooting: When your program encounters an issue or becomes unresponsive, the reset button can help you restart the program to troubleshoot problems.

1.7.4. Initialize the Sketch: The reset button is also used to start the Arduino sketch (your program) from the beginning. . It can be especially helpful when you have code that runs initialization routines during startup.

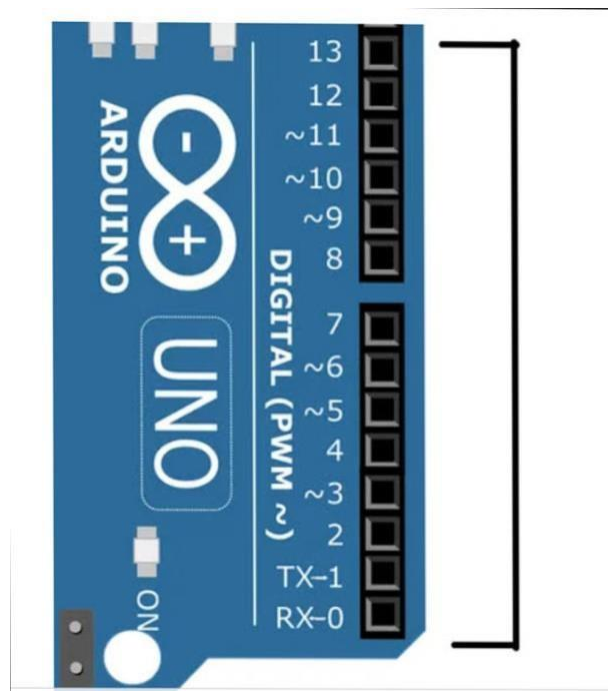
Overall, the reset button is a handy tool for controlling and debugging your Arduino projects.

1.8. DIGITAL PINS:

- Digital pins on an Arduino Uno play a crucial role in both input and output processes. These pins can be individually configured to either read input (receiving data) or provide output (sending data). Here's a breakdown of their important functions:
- **Input:** Digital pins can read incoming signals, whether it's from a sensor, switch, or any other external device. These pins are capable of detecting voltage changes (HIGH or LOW) and can interpret digital signals or binary data (1s

and 0s). This input can be used to trigger actions or make decisions within the Arduino code

- **Output:** The digital pins can generate output signals, providing power or ground to control external components. For instance, they can power LEDs, motors, or other devices based on the program instructions. By toggling the pin's state between HIGH (5 volts) and LOW (0 volts), you can control the behavior of connected components.



1.8.1 digital pins in Arduino UNO

- **PWM (Pulse Width Modulation):** Some of the digital pins on the Arduino Uno can generate PWM signals. This allows you to control the brightness of an LED, the speed of a motor, or the position of a servo motor by varying the duty cycle of the PWM signal.
- **Communication Protocols:** Digital pins can be configured for various communication protocols, such as SPI (Serial Peripheral Interface), I2C (Inter-Integrated Circuit), and UART (Universal Asynchronous Receiver-Transmitter), which are essential for interfacing with other devices, like sensors, displays, and communication modules.

Understanding the capabilities of the digital pins is fundamental for effectively interfacing with sensors, actuators, and other electronic components, allowing you to create various projects and prototypes using the Arduino Uno.

1.9 CRYSTAL OSCILLATOR : The crystal oscillator in an Arduino Uno is responsible for providing precise timing for the microcontroller. It helps regulate the clock speed and allows the microcontroller to execute instructions accurately by



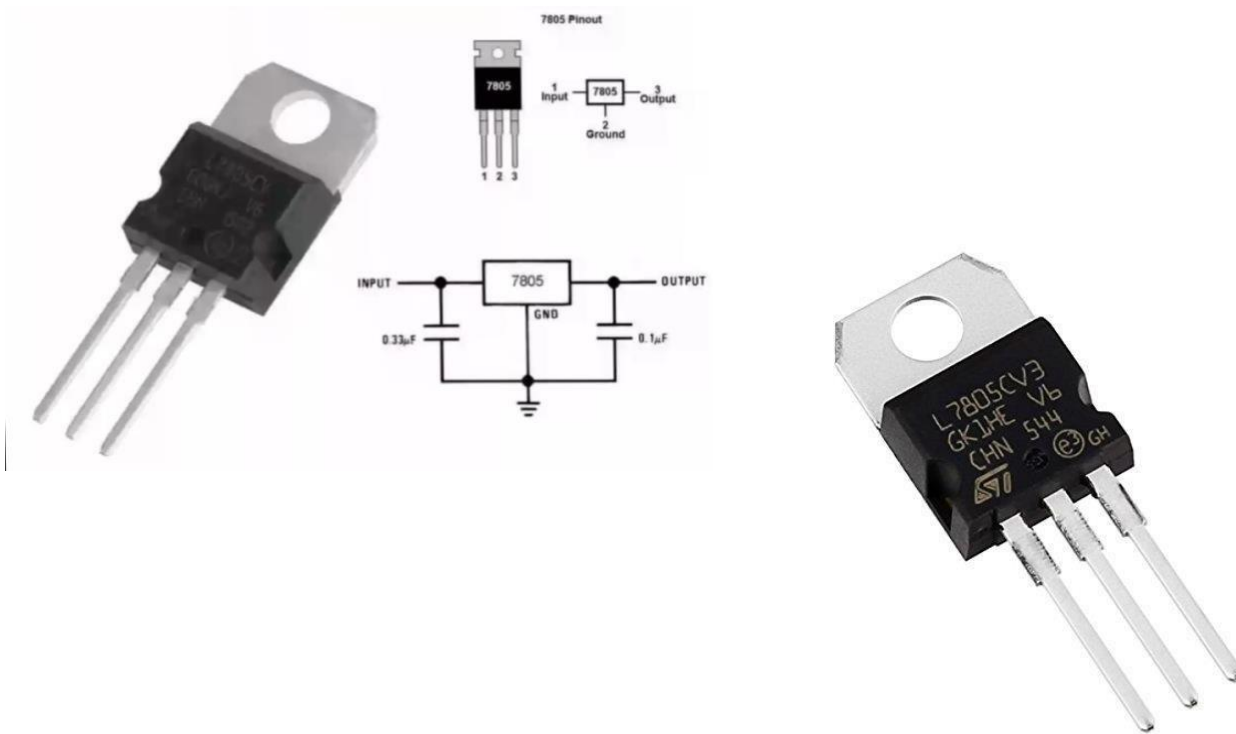
1.9.1 crystal oscillator in Arduino uno

generating a stable clock signal. This enables proper timing for various functions like timekeeping, communication, and other operations of the Arduino Uno.

- The pins generally named as XTAL1 and XTAL2, here the connection of crystal oscillator with micro-controller is given below: The reason for using two capacitors in series with crystal oscillator is to resonate with the crystal inductance which cause the crystal to oscillate on its fundamental parallel resonant mode.

1.9 VOLTAGE REGULATOR:

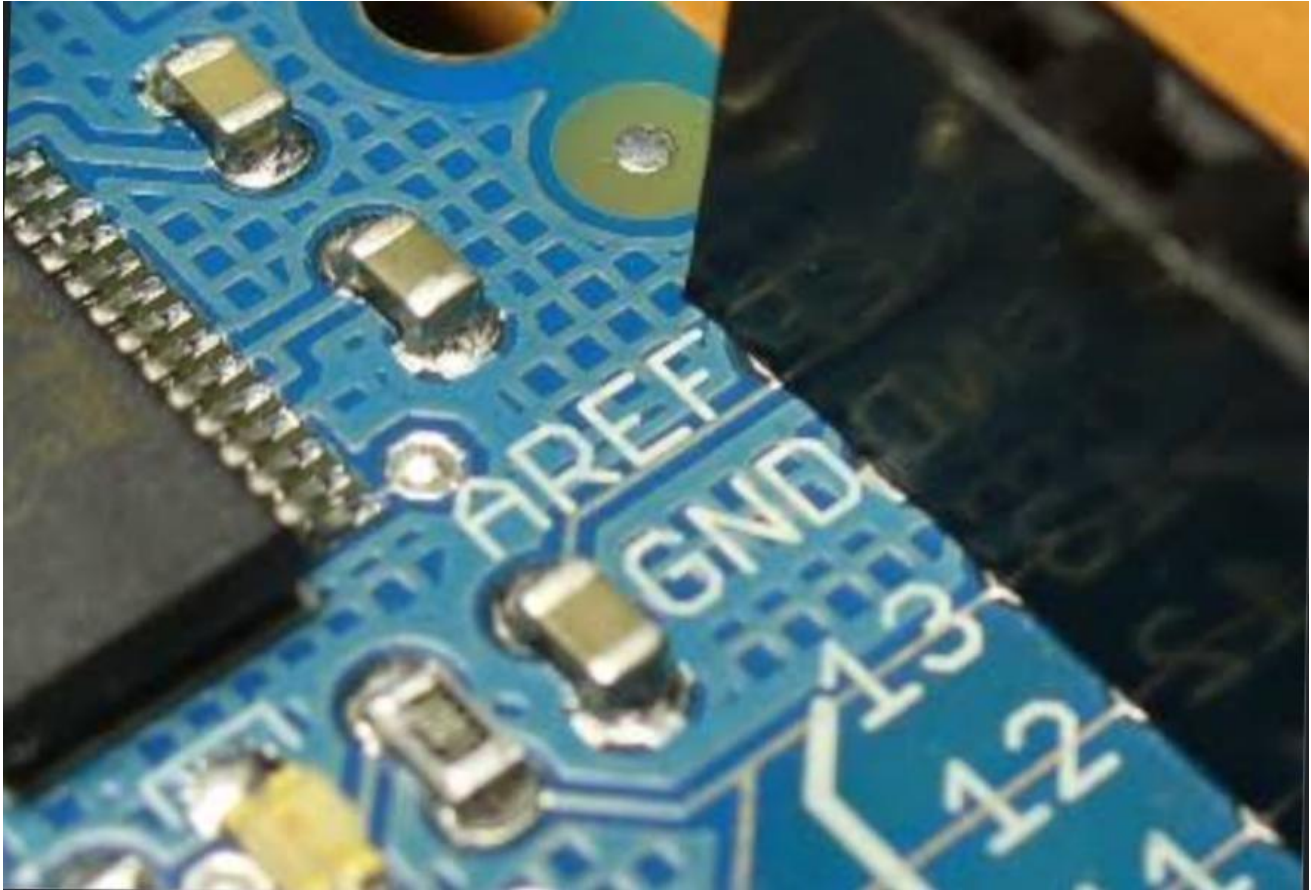
- The main role of a voltage regulator in an Arduino or any electronic circuit is to stabilize the voltage supplied to ensure a consistent and specific voltage level, enabling the proper functioning of the connected components. For an Arduino, this stability is crucial as it helps protect the board and connected devices from potential damage due to fluctuations in input voltage.



1.10.1 Voltage regulator in Arduino Uno

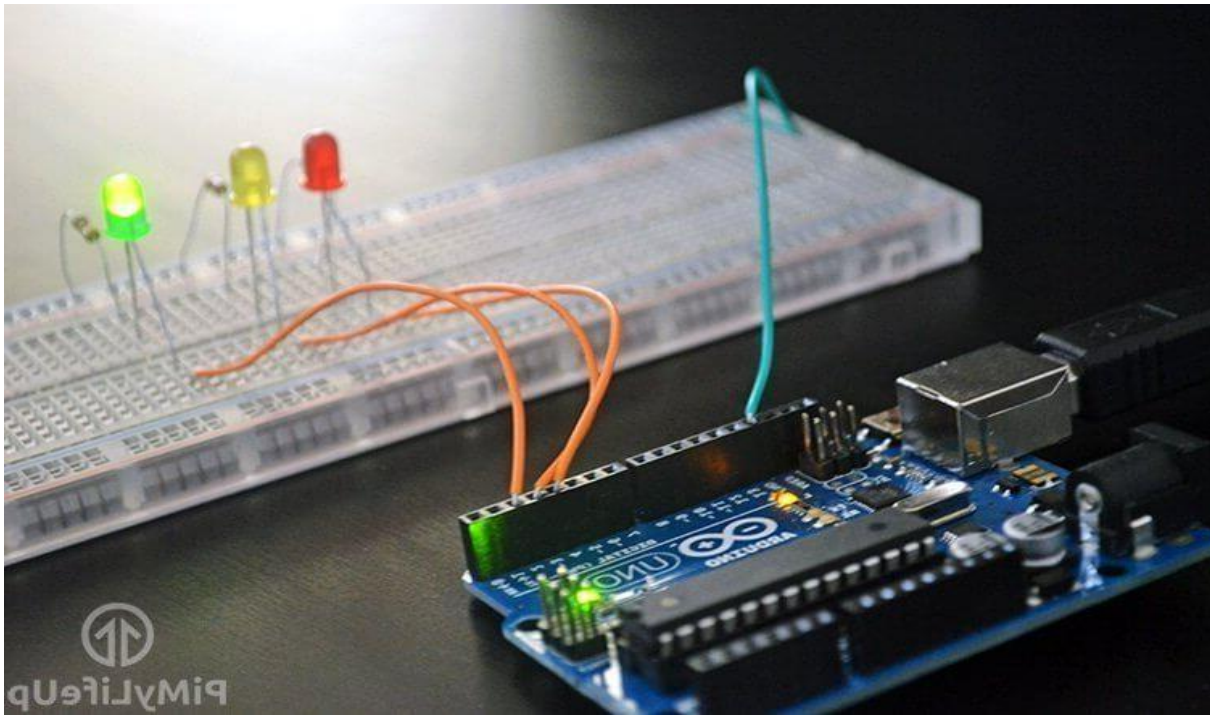
1.11 AREF (analog reference): pin on an Arduino Uno is used to set an external reference voltage for analog inputs. It allows you

to provide a more accurate reference voltage for the analog-to-digital converter (ADC). By connecting an external voltage reference to the AREF pin, you can improve the accuracy of analog readings taken by the Arduino.



1.11.1 AREF(analog reference)in Arduino uno

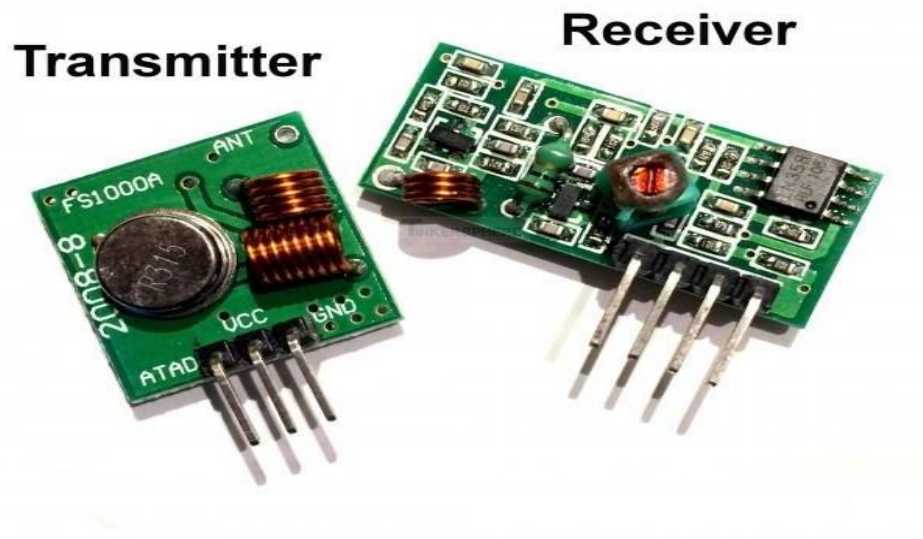
1.12 LED INDICATOR: The power LED indicator on an Arduino Uno serves as a visual indicator that board is receiving power. It lights up when the Arduino is properly connected to a power source, providing a quick visual cue that the board is getting the necessary power supply.



1.13 led indicator in Arduino Uno

1.14.1 TRANSMITTER & RECEIVER :

- The TX (transmit) and RX (receive) pins on an Arduino Uno are used for serial communication.
- The TX pin is responsible for transmitting data from the Arduino to another device, while the RX pin receives data



1.14.1 transmitter and receiver

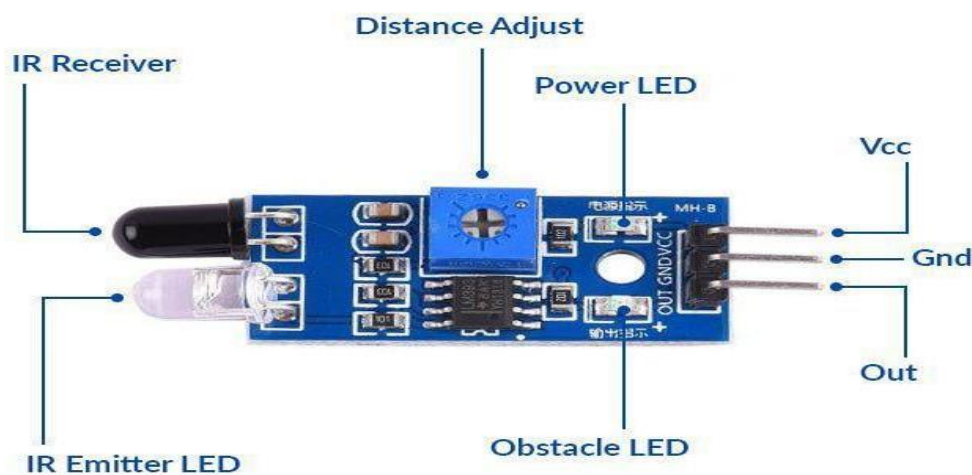
sent from an external device to the Arduino. These pins are essential for communication between the Arduino and other devices, such as sensors, displays, or computers, allowing data exchange through a serial interface.

The Arduino Uno is a popular microcontroller board known for its versatility and ease of use. Here are its key specifications:

- Microcontroller: ATmega328P.
 - Operating Voltage: 5V.
 - Input Voltage: 7-12V (recommended), 6-20V (maximum).
 - Digital I/O Pins: 14 (of which 6 provide PWM output).
 - Analog Input Pins: 6.
 - DC Current per I/O Pin: 20 mA.
 - DC Current for 3.3V Pin: 50 mA.
 - Flash Memory: 32 KB (of which 0.5 KB is used by the bootloader).
 - SRAM: 2 KB.
 - EEPROM: 1 KB.
 - Clock Speed: 16 MHz.
 - Communication: USB (for programming) and UART (serial communication).
 - In-Circuit Serial Programming (ICSP) Header.
 - Auto-reset feature for programming.
 - Power Jack for external power supply.
 - LED (built-in) on pin 13.
 - Dimensions: 68.6mm x 53.3mm.
- Weight: 25g.

2.0 IR SENSOR(infrared sensor):

- An IR sensor, or infrared sensor, is a device that detects and measures infrared radiation in its surroundings. It uses infrared light to sense nearby objects, measure heat, or receive and transmit data wirelessly.



2.0.1 pictorial diagram of IR senso

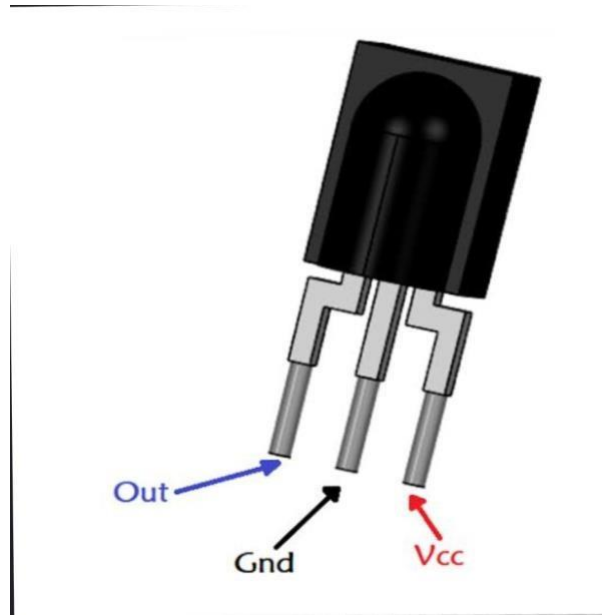
- An IR (Infrared) sensor is a device that detects and measures infrared radiation, which is not visible to the human eye but is emitted by objects as heat. IR sensors

operate based on the principle that all objects with a temperature above absolute zero emit IR radiation.

- These sensors work by detecting the IR radiation emitted by objects or by using an IR emitter and receiver to measure the reflection of emitted IR light. IR sensors find applications in various fields such as proximity sensing, motion detection, temperature measurement, remote controls, and many other technologies where the detection or measurement of IR radiation is necessary.

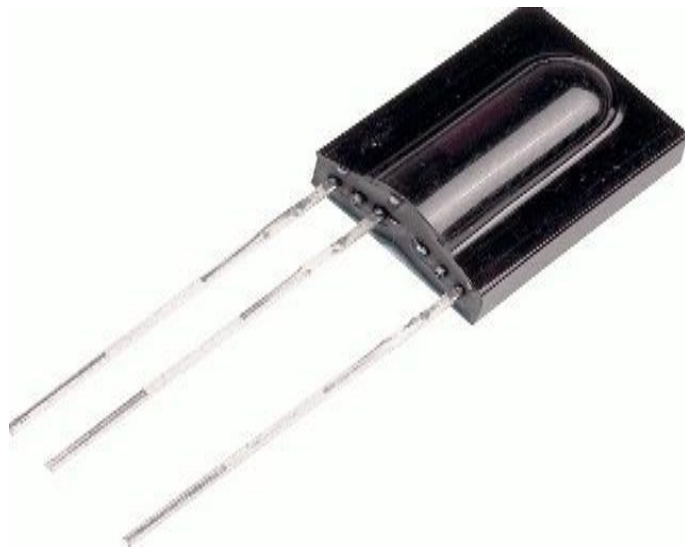
2.1 IR RECEIVER:

- The IR receiver in an IR sensor serves as the component that detects and receives infrared signals transmitted by an IR emitter or source. It plays a vital role in receiving modulated infrared light signals that convey information.
- Once the IR receiver captures these signals, it decodes and processes the information, enabling the sensor to interpret commands, data, or triggers sent through the infrared medium.
- This functionality is commonly utilized in various applications, including remote controls, proximity sensors, communication systems, and object detection devices. The



2.1.1 IR receiver in IR sensor

- IR receiver's ability to interpret and process received infrared signals is crucial for the accurate functioning and operation of the IR sensor.



2.1.2 IR receiver

3.0 IR EMITTER LED /TRANSMITTER :

- In an IR (Infrared) sensor, an IR emitter LED (Light Emitting Diode) is used to emit infrared light. This LED produces infrared light that is used to detect objects or obstacles by the IR sensor. The emitted infrared light is then reflected back or absorbed by objects, and the IR sensor detects the changes in the reflected or absorbed light, allowing the sensor to determine the presence or absence of an object.



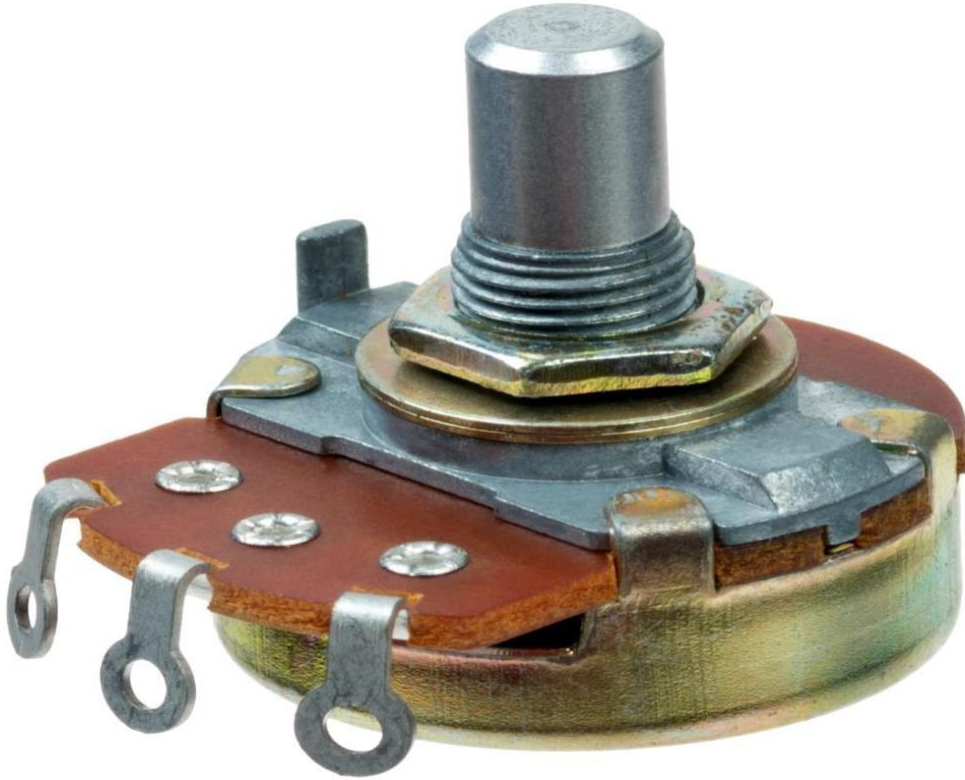
3.0.1 IR emitter LED in IR sensor

- The IR emitter LED is essential for providing the necessary infrared radiation that the sensor detects or uses to communicate, enabling functions such as object detection, proximity sensing, data transmission, or remote control operations in various devices and systems.

4.0 Power LED:

- The power LED in an IR (Infrared) sensor serves as an indicator of the sensor's operational status.
- It typically indicates that the sensor is receiving power and functioning properly.
- When the sensor is powered on, the power LED lights up, providing a visual confirmation that the sensor is active and ready to detect IR signals or perform its intended function.
- .This LED does not directly contribute to the sensor's IR detection capabilities but helps users know when the sensor is operational.

5.0 Distance adjuster potentiometer :



5.0.1 Distance adjuster potentiometer in IR sensor

- In an IR sensor, the distance adjuster potentiometer is typically used to calibrate or set the sensor's sensitivity for detecting objects at different distances. It helps fine-tune the detection range by adjusting the electrical resistance, thereby altering the sensitivity of the sensor to infrared light reflected from objects at varying distances.

6.0 Obstacle LED :

The obstacle LED in an IR (Infrared) sensor serves as an indicator that lights up when the sensor detects an obstacle or object within its range. It provides a visual signal to indicate the presence of an obstruction in front of the IR sensor.

- In an IR sensor system, you typically have both an IR LED and an IR receiver (photodiode or phototransistor)

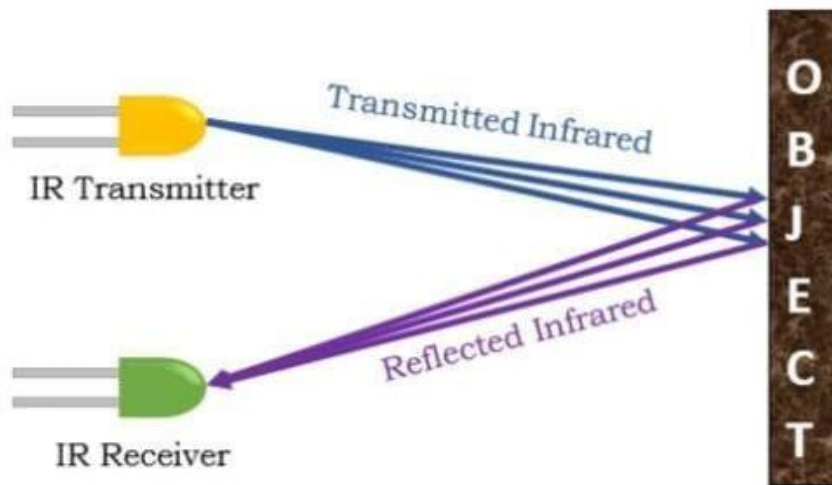
7.0 Signal LED :

- The main role of the signal LED in an IR sensor is to indicate the status of the sensor. It typically illuminates or blinks when the sensor detects infrared light, letting users know that the sensor is actively sensing or detecting an infrared signal. This LED serves as a visual indicator for the sensor's functionality.

8.0 Operation of IR sensor:

- Infrared (IR) sensors work based on the detection of infrared radiation. These sensors typically consist of an emitter that produces infrared light and a receiver that detects any reflected or emitted IR radiation. When an object or surface is in proximity to the sensor, the emitted infrared light is reflected and received by the sensor.
- The sensor then measures the intensity of the received IR radiation.

- Changes in this intensity help the sensor to determine the presence or absence of an object and its distance, based on how much infrared light is reflected back to the receiver.
- This information is converted into electrical signals, providing data about the presence or absence of an object in the sensor's vicinity.



8.0.1 IR sensor operation

APPLICATION:

- IR (infrared) sensors have various applications across different fields due to their ability to detect infrared radiation. Some common uses include:
- **Security Systems:** Infrared motion detectors are used in security systems to detect movement within a specific area, triggering alarms or activating surveillance cameras.
- **Proximity Sensors:** IR sensors can be employed in devices like smartphones and home automation systems to detect proximity, enabling features such as automatic screen dimming during calls or touchless faucets.
- **Temperature Sensing:** IR sensors can measure temperature without direct contact, used in medical devices for measuring body temperature and in industrial settings to monitor equipment and processes.
- **Object Detection:** They are used in robotics and automation to detect the presence of objects, obstacles, or individuals, enabling automated processes or preventing collisions.
- IR sensors are commonly used in applications such as proximity sensing, object detection, motion detection, and even in remote controls, where they detect infrared signals from the control device.
- They operate on the principle that objects reflect or emit infrared radiation, and the sensor can detect changes in this radiation to infer the presence or movement of objects.

Disadvantage:

- **Limited Range:** IR sensors typically have a limited detection range, especially in outdoor or open environments. They are effective for short to medium-range applications but may not work well over long distances.

3.0 SERVO MOTOR:

- The term “servo” in “servo motor” comes from the word “servomechanism,” which refers to a closed-loop control system that uses feedback to achieve and maintain a desired position, speed, or other parameters.



- Servo motors operate within such control systems,

3.0.1 Servo motor

receiving feedback from sensors to adjust their position or speed, hence the name “servo” motors.

- They “serve” or respond to the control signal and feedback to maintain the desired output.
- Servo motors are rotary actuators that allow for precise control of angular position, velocity, and acceleration.
- They are commonly used in various applications like robotics, industrial automation, remote-controlled vehicles, and more.

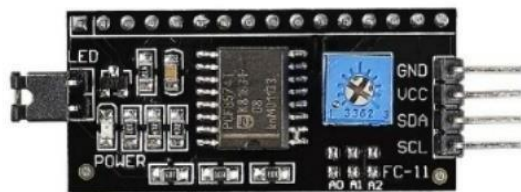
- Servos operate based on feedback control systems and typically consist of a motor, control circuit, and feedback mechanism. They provide accurate control and are popular due to their ability to maintain a specific position.

4.0 I2C module LCD display:

- An I2C module display is like a tiny screen that can show words, numbers, or symbols. It's a special screen that's easy to connect to a computer or other devices.
- You can use it to display information or messages in your projects. It's like a simple way to show things on a small

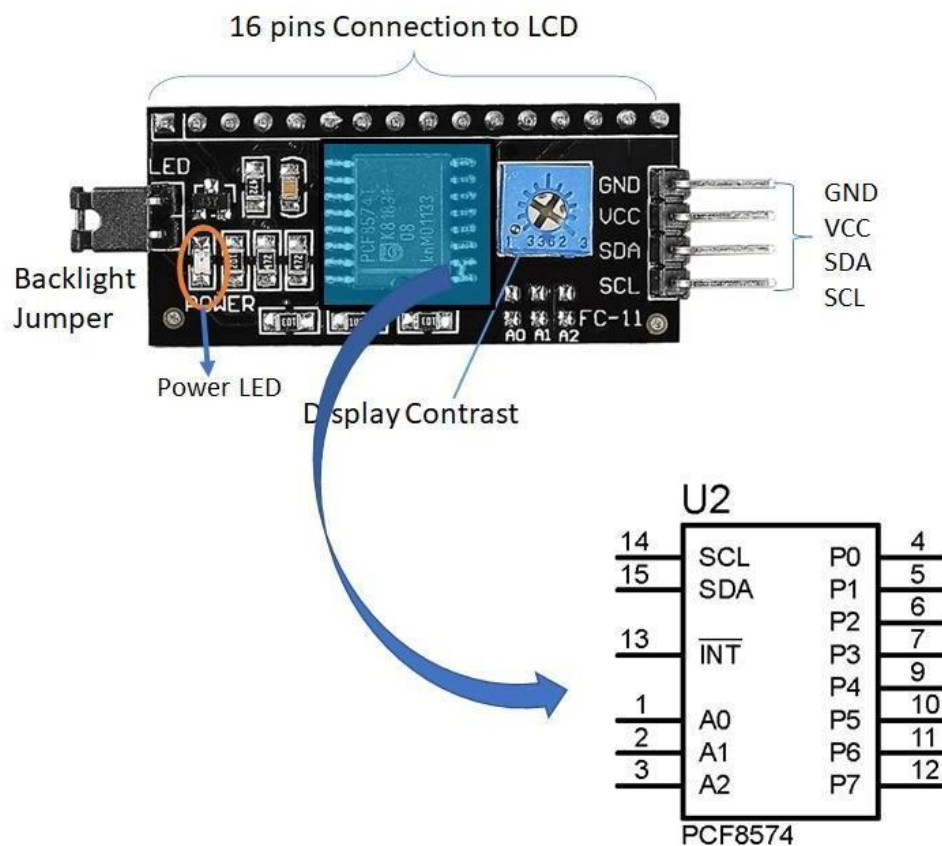
4.0.1 i2c module & lcd display

- The I2C module LCD display is a type of liquid crystal



display (LCD) that can be interfaced with a microcontroller or other electronic devices using the I2C (Inter-Integrated Circuit) communication protocol.

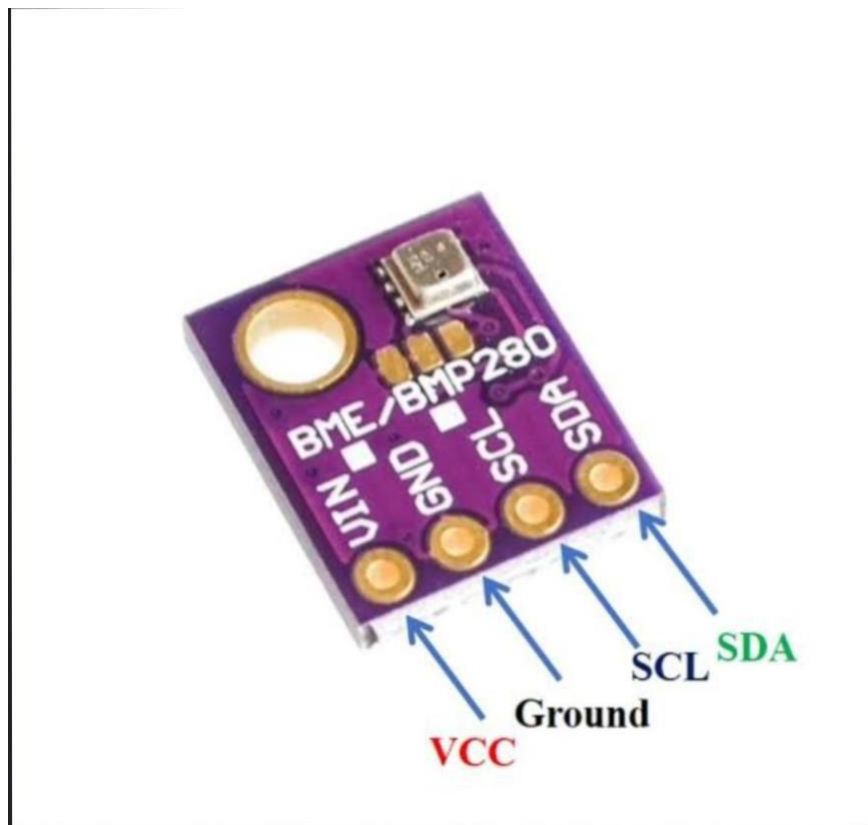
- This module typically consists of an LCD screen and an I2C backpack or module, which simplifies the process of connecting and controlling the display
- The I2C protocol allows for serial communication between



4.0.2 i2c module lcd display pin diagram

- the microcontroller and the LCD display with only two wires:

- one for data (**SDA**) and another for clock (**SCL**). This makes it easier to connect the LCD to your project, as it reduces the number of required pins and simplifies the wiring.
- **SDA**” stands for Serial Data Line. SDA is one of the two communication lines used in the I2C (Inter-Integrated Circuit) protocol, and it is responsible for transmitting data between the microcontroller and the I2C module display.
- The other line is SCL, which stands for Serial Clock Line

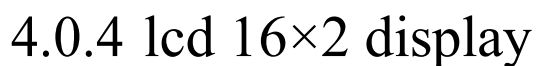


4.0.3 lcd pins

and is used to synchronize the data transfer between devices on the I2C bus.

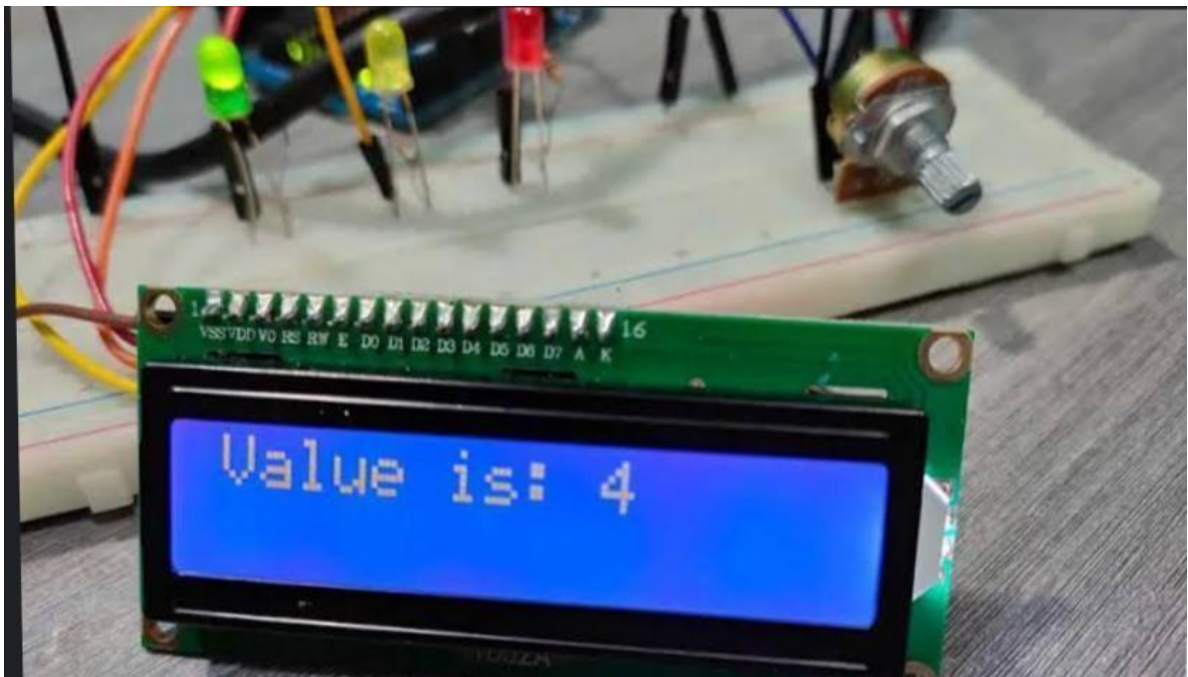
- I2C module LCD displays are commonly used in various

- They provide a convenient way to display information and text on the screen and can be controlled programmatically through the I2C interface.



- An I2C module, a “back light jumper” could refer to a hardware configuration or connection related to controlling the backlight of a display or some other component.

- This jumper might be used to enable or disable the backlight, adjust its brightness, or configure other aspects of the display.
- The main role of a backlight jumper in an I2C module, particularly for a display module like an OLED or LCD screen, is to control the backlight of the display.
- It is used to enable or disable the backlight or adjust its brightness. The backlight provides illumination to the



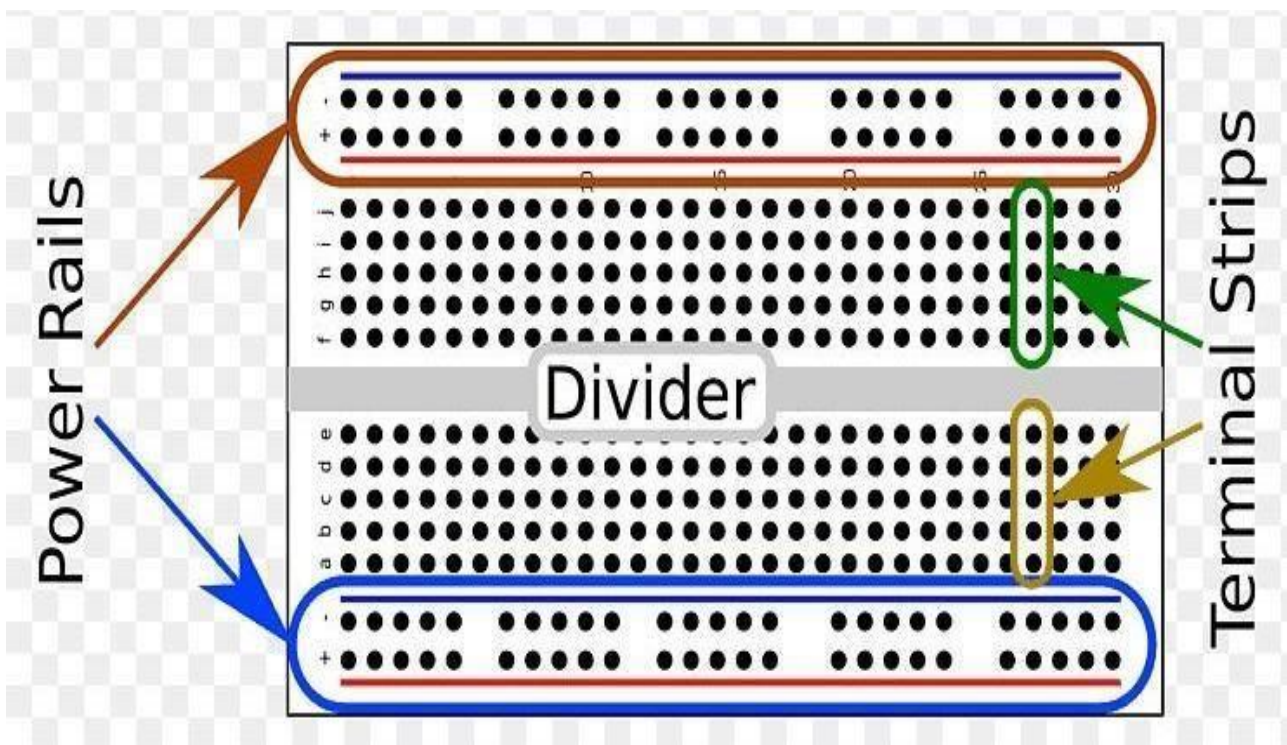
display, making the content visible.

5.0.1 icd display

- By using a jumper, you can easily control the backlight based on your application requirements.

- For example, you can configure the display to conserve power by turning off the backlight when not needed or adjust its brightness to suit different lighting conditions.
- The specific functionality and usage of the backlight jumper can vary depending on the design of the I2C module and the display it's connected to.
- It's important to refer to the module's documentation or specifications to understand how the backlight jumper should be used in your particular setup.

6.0 Bread Board :



6.0.1 Bread Board

- A breadboard is a device used for prototyping electronic circuits. It allows you to create temporary connections between electronic components without soldering. It's typically a rectangular plastic board with a grid of holes into which components and wires can be inserted to create circuits for testing and experimentation.
- Breadboards are used for prototyping and testing electronic circuits without soldering. They offer a quick and flexible way to assemble, modify, and test circuits by allowing components to be easily inserted and connected without permanent attachments. This makes it ideal for experimenting, learning, and creating temporary circuits before finalizing a design on a more permanent platform.

Breadboards are commonly used in various fields, including:

Education: Breadboards are extensively used in electronics education to teach students circuit design and experimentation.

Prototyping: Engineers, hobbyists, and professionals use breadboards to create and test new circuit designs before finalizing them for production.

Rapid Experimentation: For quick testing of electronic circuits or for creating temporary setups in labs or workshops, breadboards are an excellent choice.

Development: During the early stages of hardware development, engineers use breadboards to validate and test concepts or designs before moving to a more permanent circuit board solution.

In essence, breadboards serve as a versatile and practical tool for anyone working with electronic circuits, offering a convenient way to build, test, and modify circuits without the need for soldering.



7.0 RFID:

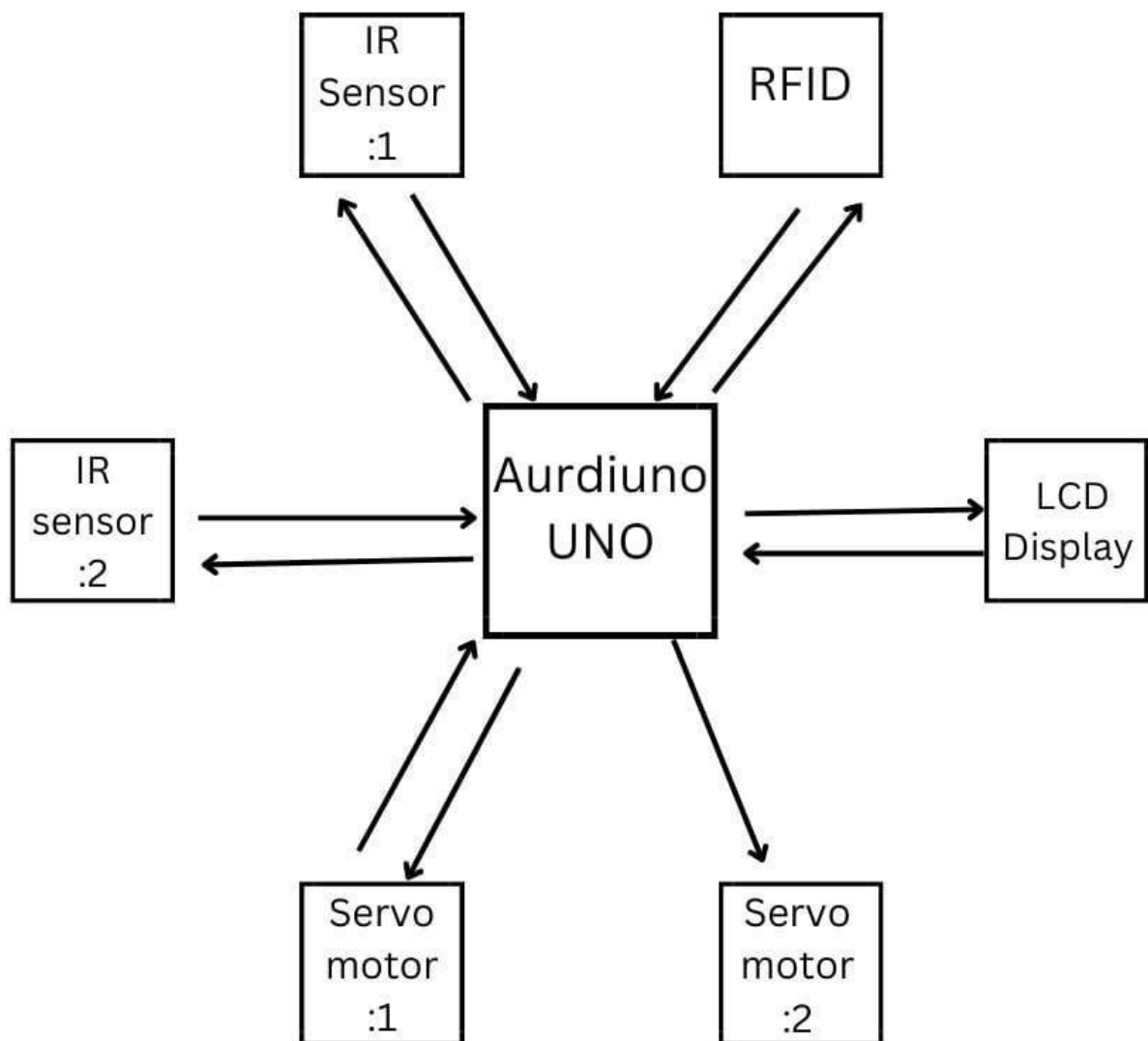
7.0.1 pictorial of RFID

- RFID (Radio-Frequency Identification) is a technology that uses electromagnetic fields to automatically identify and track tags attached to objects. These tags contain electronically stored information that can be remotely retrieved by an RFID reader.
- RFID is used in various applications such as inventory management, access control, payment systems, and more due to its ability to track items without direct contact or line-of-sight scanning.

RFID technology offers several advantages, such as:

- **Automation:** RFID enables automatic identification and tracking of objects without direct line-of-sight scanning.
- **Efficiency:** It streamlines processes like inventory management, supply chain logistics, and access control by providing quick and accurate data capture.
- **Versatility:** RFID systems can be used in various industries, including retail, healthcare, transportation, and more.
- **Reduced Errors:** It decreases human error in data collection and tracking.
- **Enhanced Security:** RFID tags can offer enhanced security features for access control, making them difficult to counterfeit.
- Overall, RFID is utilized for its ability to improve efficiency, accuracy, and security in various applications and industries.

8.0 Block diagram:



8.0.1 Block daigram

9.0 working principle :

- A movable divider using RFID, Arduino Uno, IR sensors, a servo motor, I2C module, and an LCD display can be designed to create a system that can be controlled using RFID cards. Here's a basic overview of the working principle:

1. RFID Module:

- RFID cards or tags contain unique identification information. The RFID module reads this data when a card is placed near it.

2.Arduino Uno:

The Arduino Uno acts as the main controller. It receives data from the RFID module and IR sensors and controls the servo motor, LCD display, and I2C module.

3.IR Sensors:

IR sensors are used to detect the presence of people or objects in front of the divider. They provide input to the Arduino, which can trigger the divider's movement.

4.Servo Motor:

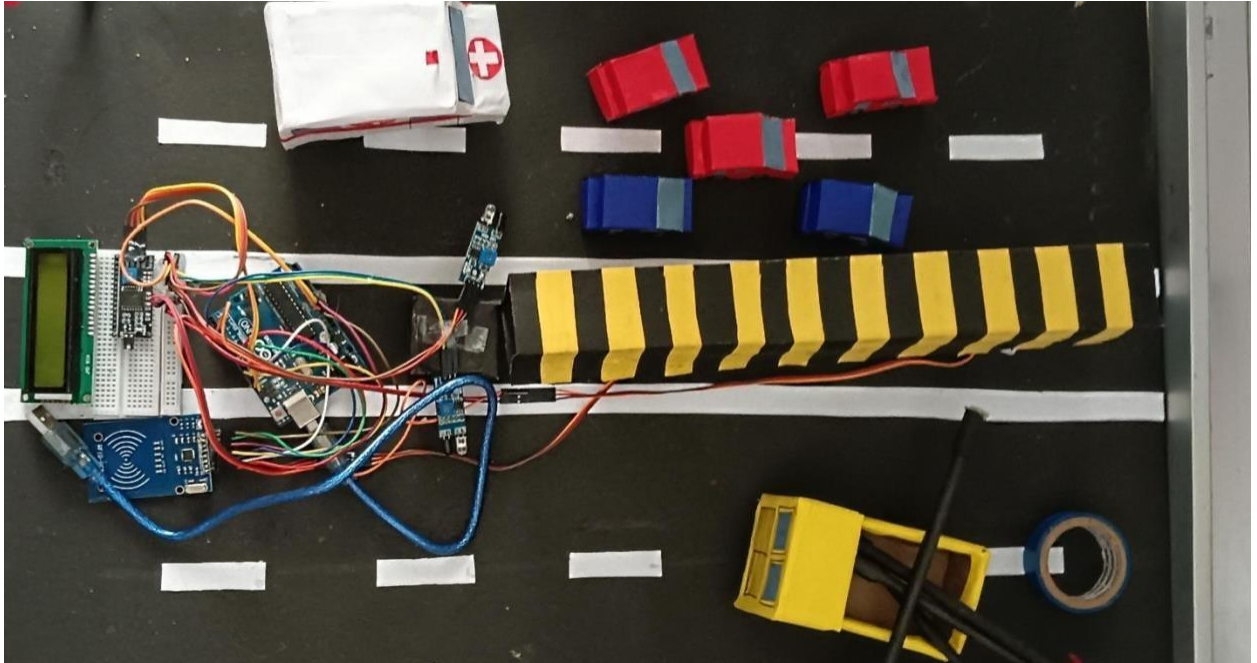
The servo motor controls the movement of the divider. It can rotate to open or close the divider based on the input from the Arduino

5.I2C Module:

The I2C module is used for connecting the LCD display to the Arduino. It simplifies the process of displaying information on the screen.

6.LCD Display:

The LCD display provides visual feedback to the user, showing information such as the status of the divider (open or closed) and other relevant data.



9.0.1 Dynamic traffic flow management

Working Principle:

- When an RFID card is presented to the RFID module, it reads the card's data and sends it to the Arduino Uno.
- The Arduino checks the received RFID data to determine if the cardholder has permission to control the divider.
- If access is granted, the Arduino may check the input from IR sensors to ensure no one is obstructing the divider's path.
- Based on the RFID verification and IR sensor input, the Arduino sends commands to the servo motor to either open or close the movable divider.
- The Arduino also sends relevant information to the LCD display through the I2C module, providing real-time status updates.
- The system continuously monitors RFID and IR inputs, allowing authorized users to control the divider's movement and displaying its status on the LCD.
- This system ensures controlled access and safety by combining RFID authentication with obstacle detection using IR sensors and providing visual feedback through the LCD display. The Arduino Uno acts as the central brain of

the system, orchestrating all these components to create a functional movable divider.

Program:

```
*/  
  
#include <Servo.h>  
#include <SPI.h>  
#include <MFRC522.h>  
#include <Wire.h>  
#include <LiquidCrystal_I2C.h>  
LiquidCrystal_I2C lcd(0x27, 16, 2);  
  
Servo myservo;  
Servo myservo2;  
  
#define          9      // Configurable, see typical pin  
RST_PIN layout  
above  
#define SS_PIN    10    // Configurable, see typical pin  
layout above  
#define ACCESS_DELAY 2000  
#define DENIED_DELAY 1000
```



```
MFRC522 mfrc522(SS_PIN, RST_PIN); // Create MFRC522
instance
```

```
Void setup() {
```

```
    Serial.begin(9600);    // Initialize serial communications
with the PC
```

```
    While (!Serial);    // Do nothing if no serial port is
opened (added for Arduinos based on ATMEGA32U4)
```

```
    SPI.begin();    // Init SPI bus
```

```
    Mfrc522.PCD_Init();    // Init
```

```
    MFRC522
```

```
    Delay(4);    // Optional delay. Some board do
need more time after init to be ready, see Readme
```

```
    Mfrc522.PCD_DumpVersionToSerial();    // Show
details of PCD – MFRC522 Card Reader details
```

```
    Serial.println(F("Scan PICC to see UID, SAK, type, and data
blocks..."));
```

```
    Lcd.init();
```

```
    Lcd.backlight();
```

```
    pinMode(3,INPUT);//ir1
```

```
    pinMode(4,INPUT);//ir2
```

```
pinMode(7,OUTPUT
);
pinMode(8,OUTPUT
); lcd.setCursor(0,0);
lcd.println("MOVABLE ROAD");
lcd.setCursor(0,1);
lcd.print("DIVIDER"
); delay(2000);
lcd.clear();
```

```
myservo2.attach(2);
myservo.attach(6);//attaches the servo on pin 9 to the servo
object
myservo.write(0);
myservo2.write(0);
delay(500);
myservo.write(180);
myservo2.write(180);
delay(500);
myservo.write(90);
myservo2.write(90);
```

```

Serial.println("Put your card to the reader...");
Serial.println();
}

Void loop() {
    // Reset the loop if no new card present on the
    sensor/reader. This saves the entire process when idle.
    If ( ! mfrc522.PICC_IsNewCardPresent())
        { Return;
        }

    // Select one of the cards
    If ( ! mfrc522.PICC_ReadCardSerial()) {
        Return;
    }

    // Dump debug info about the card; PICC_HaltA() is
    automatically called
    Mfrc522.PICC_DumpToSerial(&(mfrc522.uid));
    Delay(1000);
}

```

```

Serial.println();
Serial.println();
  Serial.print("UID tag :");
String content= "";
Byte letter;
For (byte I = 0; I < mfrc522.uid.size; i++)
{
  Serial.print(mfrc522.uid.uidByte[i] < 0x10 ? " 0" : " ");
  Serial.print(mfrc522.uid.uidByte[i], HEX);
  Content.concat(String(mfrc522.uid.uidByte[i] < 0x10 ? " 0" :
" "));
  Content.concat(String(mfrc522.uid.uidByte[i], HEX));
}
Serial.println();
Serial.print("Message : ");
  Content.toUpperCase();

  If (content.substring(1) == "C9 E3 7C B2") //change here the
UID of the card/cards that you want to give access
{
  Serial.println("Authorized access");
}

```

```
Serial.println();
digitalWrite(8, LOW);
digitalWrite(7,HIGH);//GREEN LED
chk();
lcd.setCursor(0,0);
lcd.print("MOVE FORWARD");
lcd.clear();
delay(1000);
lcd.print("AUTO LOCKING IN ");
lcd.setCursor(0,1);
lcd.println("  5 SEC
"); delay(5000);
digitalWrite(8,HIGH);//RED LED
digitalWrite(7,LOW);//GREEN LED
}
```

```
Else {
    Serial.println(" Access denied");
    digitalWrite(8, HIGH);
    lcd.print("ACCESS DENIED");
}
```

```

    delay(1000);
}
Lcd.clear();
}
Void chk()
{
    Int
    a=digitalRead(3);
    Int
    b=digitalRead(4);
    Serial.print("a=");
    Serial.print(a);
    Serial.print("b=");
    Serial.print(b);
    If(a==0 && b==0)
    {
        Myservo.write(90);
        Myservo2.write(90);
        Lcd.print("DON'T MOVE");
        Delay(4000);

    }
}

```

```
Else if(a==1&&b==0)
{
  Myservo.write(0);
  Myservo2.write(0);
  Lcd.print("DIV MOVES LEFT");
  Delay(5000);
}
Else if(a==0&&b==1)
{
  Myservo.write(180);
  Myservo2.write(180);
  Lcd.print("MOVES RIGHT");
  Delay(5000);
}
Else if (a==1&&b==1)
{
  Myservo.write(90);
  Myservo2.write(90);
  Lcd.print("DO NOT MOVE");
  Delay(5000);
}
```

```
}  
Myservo.write(90);  
  Myservo2.write(90);  
}
```

Advantages:

1. no delay time
2. saves the human life
3. Emergency vehicles may enter ,not other vehicles
4. Death cases may reduce

Disadvantage :

1. it may not used at 4 way junction.
2. Complexity

Applications :

Emergency vehicles like:

1. ambulance
2. Fire extinguisher

CONCLUSION :

- Dynamic Traffic Flow Management” that uses IR sensors to detect traffic and LCD displays to ensure the safety of users in danger by allowing them to go through temporary dividers using RFID cards that are only given to cardholders and emergency vehicles like ambulances and fire extinguishers.

FUTURESCOPE :

- About 30% of deaths are caused due to delayed ambulance.
- 50% of heart attack cases reach hospital late , which can constitute unavailability of ambulances too but majority of it is due to patients stuck in.
- The future scope of this project is constructing a movable divider which gives path to the ambulance and other emergency vehicles got stuck in traffic. by this we can reduce the ambulance delay and save the lives of patients who got strucked in the ambulances.

